

1 When Is Consumption Growth Declining in m ?

Figure ?? depicts the expected consumption growth factor as a strictly declining function of the cash-on-hand ratio. To investigate this, define

$$\Upsilon(m_t) \equiv \mathcal{G}_{t+1}c(\tilde{\mathcal{R}}_{t+1}a(m_t) + \xi_{t+1})/c(m_t) = c_{t+1}/c_t$$

and the proposition in which we are interested is

$$(d/dm_t)\underbrace{\mathbb{E}_t[\Upsilon(m_t)]}_{\equiv \Upsilon_{t+1}} < 0$$

or differentiating through the expectations operator, what we want is

$$\mathbb{E}_t \left[\mathcal{G}_{t+1} \left(\frac{c'(m_{t+1})\tilde{\mathcal{R}}_{t+1}a'(m_t)c(m_t) - c(m_{t+1})c'(m_t)}{c(m_t)^2} \right) \right] < 0. \quad (1)$$

Henceforth indicating appropriate arguments by the corresponding subscript (e.g. $c'_{t+1} \equiv c'(m_{t+1})$), since $\mathcal{G}_{t+1}\tilde{\mathcal{R}}_{t+1} = R$, the portion of the LHS of equation (1) in brackets can be manipulated to yield

$$\begin{aligned} c_t \Upsilon'_{t+1} &= c'_{t+1}a'_t R - c'_t \mathcal{G}_{t+1}c_{t+1}/c_t \\ &= c'_{t+1}a'_t R - c'_t \Upsilon_{t+1}. \end{aligned}$$

Now differentiate the Euler equation with respect to m_t :

$$\begin{aligned} 1 &= R\beta \mathbb{E}_t[\Upsilon_{t+1}^{-\gamma}] \\ 0 &= \mathbb{E}_t[\Upsilon_{t+1}^{-\gamma-1} \Upsilon'_{t+1}] \\ &= \mathbb{E}_t[\Upsilon_{t+1}^{-\gamma-1}] \mathbb{E}_t[\Upsilon'_{t+1}] + \text{cov}_t(\Upsilon_{t+1}^{-\gamma-1}, \Upsilon'_{t+1}) \\ \mathbb{E}_t[\Upsilon'_{t+1}] &= -\text{cov}_t(\Upsilon_{t+1}^{-\gamma-1}, \Upsilon'_{t+1})/\mathbb{E}_t[\Upsilon_{t+1}^{-\gamma-1}] \end{aligned} \quad (2)$$

but since $\Upsilon_{t+1} > 0$ we can see from (2) that (1) is equivalent to

$$\text{cov}_t(\Upsilon_{t+1}^{-\gamma-1}, \Upsilon'_{t+1}) > 0$$

which, using (2), will be true if

$$\text{cov}_t(\Upsilon_{t+1}^{-\gamma-1}, c'_{t+1}a'_t R - c'_t \Upsilon_{t+1}) > 0$$

which in turn will be true if both

$$\text{cov}_t(\Upsilon_{t+1}^{-\gamma-1}, c'_{t+1}) > 0$$

and

$$\text{cov}_t(\Upsilon_{t+1}^{-\gamma-1}, \Upsilon_{t+1}) < 0.$$

The latter proposition is obviously true under our assumption $\gamma > 1$. The former will be true if

$$\text{cov}_t((\mathcal{G}\psi_{t+1}c(m_{t+1}))^{-\gamma-1}, c'(m_{t+1})) > 0.$$

The two shocks cause two kinds of variation in m_{t+1} . Variations due to ξ_{t+1} satisfy the proposition, since a higher draw of ξ both reduces $c_{t+1}^{-\gamma-1}$ and reduces the marginal propensity to consume. However, permanent shocks have conflicting effects. On the one hand, a higher draw of ψ_{t+1} will reduce m_{t+1} , thus increasing both $c_{t+1}^{-\gamma-1}$ and c'_{t+1} . On the other hand, the $c_{t+1}^{-\gamma-1}$ term is multiplied by $\mathcal{G}\psi_{t+1}$, so the effect of a higher ψ_{t+1} could be to decrease the first term in the covariance, leading to a negative covariance with the second term. (Analogously, a lower permanent shock ψ_{t+1} can also lead a negative correlation.)